

ROCKINGHAM COUNTY SHILOH AP, NC, USA

WMO: 742077

Lat: 36.437N

Lon: 79.851W

Elev: 211

StdP: 98.81

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 03763

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -6.7	(d) -22.3	(e) 0.5	(f) -2.9	(g) -18.2	(h) 0.8	(i) -2.0	(j) 7.4	(k) 10.0	(l) 6.7	(m) 9.5	(n) 0.5	(o) 250	(p) 0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	34.0	23.6	32.8	23.2	31.8	22.9	25.5	31.2	24.9	30.3	24.2	29.2	2.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.4	27.8	23.4	18.7	27.0	22.8	18.0	26.3	79.4	31.5	76.6	30.3	74.1	29.3	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.3	(b) 5.2	(c) 4.5	(e) -14.7	(f) 36.3	(g) 4.3	(h) 1.5	(i) -17.8	(j) 37.4	(k) -20.3	(l) 38.2	(m) -22.7	(n) 39.1	(o) -25.9	(p) 40.1		
(a) 6.3	(b) 5.2	(c) 4.5	(e) -15.1	(f) 26.4	(g) 4.1	(h) 0.9	(i) -18.1	(j) 27.1	(k) -20.5	(l) 27.6	(m) -22.8	(n) 28.1	(o) -25.8	(p) 28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	3.3	5.2	9.3	14.6	19.5	23.4	25.0	24.2	21.0	14.7	8.5	5.8
	DBStd	8.76	5.43	5.44	5.47	4.67	3.92	2.82	2.38	2.65	3.27	4.43	4.61	4.68
	HDD10.0	703	218	153	82	13	0	0	0	0	0	10	81	147
	HDD18.3	2112	467	369	283	127	37	2	0	1	14	129	294	388
	CDD10.0	2379	9	18	61	152	294	402	465	439	329	156	37	17
	CDD18.3	746	0	1	2	17	73	154	207	182	94	17	1	1
	CDH23.3	6901	0	3	46	248	686	1484	1920	1585	753	163	9	3
	CDH26.7	2707	0	0	3	61	215	606	861	658	272	31	1	0
Wind	WSAvg	1.3	1.5	1.8	1.8	1.9	1.2	1.0	0.9	0.8	0.9	1.1	1.2	1.3
Precipitation	PrecAvg	1131	91	75	103	93	92	105	110	106	113	81	77	85
	PrecMax	1590	184	136	226	162	200	309	195	218	311	284	222	162
	PrecMin	798	21	29	21	14	16	21	42	23	15	0	4	22
	PrecStd	169	39	28	46	41	47	61	36	42	70	62	51	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	23.3	27.1	31.3	32.7	35.2	36.3	35.8	33.0	29.7	24.9	22.6
		MCWB	14.3	16.0	15.6	17.4	21.6	23.8	24.1	22.0	21.2	20.7	14.3	15.7
	2%	DB	17.7	20.2	24.7	28.4	30.7	33.1	34.1	33.6	31.7	27.3	21.6	18.9
		MCWB	12.7	13.6	15.1	17.6	21.2	23.6	24.0	23.4	21.6	18.6	14.8	15.0
	5%	DB	15.1	17.7	22.1	26.4	29.1	32.1	32.8	32.1	30.1	25.1	19.6	16.6
		MCWB	10.0	11.2	13.6	16.7	20.9	23.2	23.8	23.2	21.7	18.1	14.0	13.0
	10%	DB	12.3	14.8	19.4	24.0	27.5	30.4	31.4	30.4	27.9	22.9	17.4	14.0
		MCWB	8.6	8.7	12.5	15.8	20.2	22.6	23.4	22.8	20.9	17.8	12.6	10.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	17.8	18.2	21.0	23.9	25.8	26.5	26.1	24.6	22.8	19.2	18.6
		MCDB	18.5	21.9	23.7	25.6	29.3	32.0	32.2	31.5	29.2	25.3	21.1	20.8
	2%	WB	14.0	15.9	16.7	19.5	22.8	25.0	25.7	25.3	23.8	21.5	17.2	16.2
		MCDB	16.7	18.4	21.3	24.4	27.9	31.2	31.5	30.2	28.5	24.3	19.6	17.5
	5%	WB	11.4	12.9	15.5	18.3	22.0	24.2	25.1	24.5	23.0	20.3	15.2	13.5
		MCDB	13.7	16.5	20.0	23.6	26.9	30.0	30.7	29.0	27.2	23.4	18.1	15.8
	10%	WB	8.7	9.7	14.0	17.2	21.2	23.5	24.4	23.8	22.1	18.6	13.1	10.9
		MCDB	11.9	12.9	17.5	22.2	25.7	28.7	29.5	27.8	25.4	22.0	16.3	13.2
Mean Daily Temperature Range	5% DB	MDBR	11.6	12.1	13.4	14.3	12.9	12.8	12.0	11.6	11.6	13.6	13.5	11.2
		MCDBR	15.1	15.5	16.6	17.1	14.9	13.9	13.7	13.6	14.7	15.5	16.3	14.0
		MCWBR	10.1	9.7	8.4	7.4	6.6	5.6	4.9	4.9	6.1	8.0	9.6	9.5
	5% WB	MCDBR	12.3	13.5	13.2	13.4	12.4	12.8	12.2	11.3	12.1	11.4	12.1	11.2
		MCWBR	10.2	10.2	7.5	6.8	5.9	5.7	5.0	4.8	5.6	6.5	8.7	9.0
Clear-Sky Solar Irradiance	taub	0.304	0.320	0.349	0.395	0.435	0.478	0.516	0.513	0.410	0.365	0.320	0.307	
	taud	2.512	2.460	2.425	2.302	2.214	2.132	2.051	2.039	2.325	2.434	2.517	2.538	
	Ebn at Noon	896	919	919	887	853	813	778	771	843	855	868	869	
	Edh at Noon	80	95	108	129	143	155	167	166	118	97	79	74	
All-Sky Solar Radiation	RadAvg	2.50	3.16	4.21	5.26	5.81	6.28	5.84	5.34	4.44	3.62	2.75	2.13	
	RadStd	0.17	0.38	0.28	0.42	0.48	0.46	0.39	0.28	0.42	0.45	0.22	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+141	+92

Nomenclature: See separate page